Circular Permutations

Arrangement of objects over a closed path is called a circular permutation.

The no. of circular permutations of n objects is $(n-1)!$

Note ① Among n objects take one of the object and fix the position. Then remaining $(n-1)$ objects are to be arranged in $(n-1)!$ ways.

② The no. of circular permutations of r objects from a collection of n objects is $\frac{1}{r} {}^{n}P_{r}$.

① (a) In how many ways can seven people be arranged about a circular table?

(b) If two of the people insist on sitting next to each other, how many arrangements are possible?

▶ (a) No. of ways to arrange 7 people over a circular table $= (7-1)! = 6!$.

(b) Now consider two persons together as one person and these two persons can be arranged among themselves in $2!$ ways.

Total no. of persons $= 7 - 2 + 1 = 6$

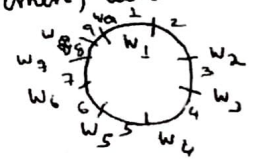
No. of ways to arrange 2 among 7 people together over a circular table $= (6-1)! \times 2! = 5! \times 2!$

② Sixteen people are to be seated at two circular tables, one of which seats 10 while the other seats 6. How many different seating arrangements are possible?

▶ We can select the 10 people to be seated at the table for 10 in ${}^{16}C_{10}$ ways. For each such selection there $9!$ ways of arranging the 10 people around the table. The remaining 6 people can be seated the other table in $5!$ ways. Consequently, there are ${}^{16}C_{10} \times 9! \times 5!$ ways to seat the 16 people around the two given tables.

③ A committee of 15, nine women and six men is to be seated at a circular table with 15 seats. In how many ways can the seats to be assigned so that no two men are seated next to each other?

▶ The 9 women can be seated around the table in $8!$ ways. Each such arrangements provides 9 spaces (b/w women) where a man can be placed. We can select six of these places to accomodate 6 men in ${}^{9}C_{6} \times 6!$ ways.



Consequently, the no. of arrangements under the given condition is $= 8! \times {}^{9}C_{6} \times 6!$

Combinations

Combination mainly deals with Selection. The no. of ways in which r objects can be selected from a collection of 'n' distinct objects is denoted by ${}^nC_r$ (or) $C(n,r)$ (or) $\binom{n}{r}$ and is given by

$$ {}^nC_r = \frac{n!}{(n-r)!\,r!} = \frac{{}^nP_r}{r!} $$

Note: ① ${}^nC_r = {}^nC_{n-r}$ ② ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

① Evaluate each of the following:

ⓐ $C(10,4)$ ⓑ $\binom{12}{7}$ ⓒ $C(14,12)$ ⓓ $\binom{15}{10}$

▲ ⓐ $C(10,4) = \frac{10!}{6!\,4!}$ ⓑ $\binom{12}{7} = \frac{12!}{5!\,7!}$

ⓒ $C(14,12) = \frac{14!}{2!\,12!}$ ⓓ $\binom{15}{10} = \frac{15!}{5!\,10!}$

② A student must answer 5 out of 10 questions in a test including at least 3 from the last five questions. In how many ways can a student answer 5 questions?

▲ Out of 10 questions 5 must be answered.

First Five questions	Last Five questions
$2 \to {}^5C_2$	$3 \to {}^5C_3$
$1 \to {}^5C_1$	$4 \to {}^5C_4$
$0 \to {}^5C_0$	$5 \to {}^5C_5$

∴ No. of always to answer $= [{}^5C_2 \times {}^5C_3] + [{}^5C_1 \times {}^5C_4] + [{}^5C_0 \times {}^5C_5]$

③ A committee of 12 is to be selected from 10 men and 10 women. In how many ways can the selection be carried out if ⓐ there are no restrictions ⓑ there must be six men and six women ⓒ there must be an even no. of women? ⓓ there must be more women than men? ⓔ there must be at least 8 men?

▲ ⓐ ${}^{20}C_{12}$

ⓑ ${}^{10}C_6 \times {}^{10}C_6$

ⓒ

Men	Women
$10 \to {}^{10}C_{10}$	$2 \to {}^{10}C_2$
$8 \to {}^{10}C_8$	$4 \to {}^{10}C_4$
$6 \to {}^{10}C_6$	$6 \to {}^{10}C_6$
$4 \to {}^{10}C_4$	$8 \to {}^{10}C_8$
$2 \to {}^{10}C_2$	$10 \to {}^{10}C_{10}$

∴ No. of ways to form the committee of 12 with an even no. of women

$= [{}^{10}C_{10} \times {}^{10}C_2] + [{}^{10}C_8 \times {}^{10}C_4] + [{}^{10}C_6 \times {}^{10}C_6] + [{}^{10}C_4 \times {}^{10}C_8] + [{}^{10}C_2 \times {}^{10}C_{10}]$

(d).

Men	Women
$5 \to {}^{10}C_5$	$7 \to {}^{10}C_7$
$4 \to {}^{10}C_4$	$8 \to {}^{10}C_8$
$3 \to {}^{10}C_3$	$9 \to {}^{10}C_9$
$2 \to {}^{10}C_{10}$	$10 \to {}^{10}C_{10}$

∴ No. of ways to form the Committee of 12 with more women than men

$= [{}^{10}C_5 \times {}^{10}C_7] + [{}^{10}C_4 \times {}^{10}C_8] + [{}^{10}C_3 \times {}^{10}C_9] + [{}^{10}C_2 \times {}^{10}C_{10}]$

(e).

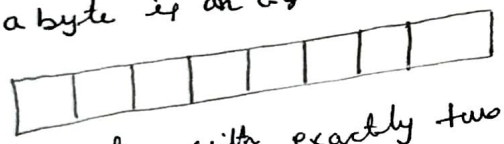
Men	Women
$8 \to {}^{10}C_8$	$4 \to {}^{10}C_4$
$9 \to {}^{10}C_9$	$3 \to {}^{10}C_3$
$10 \to {}^{10}C_{10}$	$2 \to {}^{10}C_2$

∴ No. of ways to form the Committee of 12 with at least 8 men

$= [{}^{10}C_8 \times {}^{10}C_4] + [{}^{10}C_9 \times {}^{10}C_3] + [{}^{10}C_{10} \times {}^{10}C_2]$

(4). How many bytes contain (a) exactly two 1's (b) exactly four 1's (c) exactly six 1's (d) at least six 1's.

Since, a byte is an Eight bit sequence.



(a) The no. of bytes with exactly two 1's among 8 places can be selected in ${}^8C_2$ ways.

(b). —11— Four 1's

in ${}^8C_4$ ways.

(c) —11— Six 1's

in ${}^8C_6$ ways.

(d) The no. of bytes with at least Six 1's is

${}^8C_6 + {}^8C_7 + {}^8C_8$

(5). A student is to answer 7 out of 10 questions on an examination. In how many ways can he make his selection if (a) there are no restrictions? (b) he must answer the first two questions? (c) he must answer at least 4 of the first six questions?

(a) ${}^{10}C_7$ ways (b) If he answer first two questions, then he has to answer 5 out 8 and this can be done in ${}^8C_5$ ways.

(c).

First six questions

$4 \to {}^6C_4$
$5 \to {}^6C_5$
$6 \to {}^6C_6$

Remaining 4 questions

$3 \to {}^4C_3$
$2 \to {}^4C_2$
$1 \to {}^4C_1$

$\therefore [{}^6C_4 \times {}^4C_3] + [{}^6C_5 \times {}^4C_2] + [{}^6C_6 \times {}^4C_1] =$

Consequently, the $12!$ arrangements $= \frac{12!}{[4!]^2[2!]^2}$

⑨ How many arrangements of the letters in "MISSISSIPPI" have no consecutive S's?

▲ $-M-I-I-I-P-P-I-$

there are $\frac{7!}{4!2!}$ arrangements without 'S'.

There 8 possible places for S and that can be selected in $8C_4$ ways.

Thus, there are $\frac{7!}{4!2!} \times 8C_4$ arrangements are available with no consecutive S's?

How many arrangements of "EFFLORESCENCE" has consecutive C's and consecutive F's but no consecutive E's?

We have: E → 4, F → 2, C → 2, L → 1, O → 1, R → 1, N → 1, S → 1

If 2 C's and 2 F's are together, then we have the following letters:

E → 4, F → 1, C → 1, L → 1, O → 1, R → 1, N → 1, S → 1

without 4 E's, we have:

- F - C - L - O - R - N - S -

There are 7! arrangements without E's.

There are 8 possible placed for E and that can be selected in $8C_4$ ways.

Thus, there are $7! \times 8C_4$ arrangements are there with consecutive C's and consecutive F's but no consecutive E's.

9. For the strings of length 10 made up from 0, 1, and 2, how many have

(a) Even

In coding theory, a string consists of the digits 0, 1, and 2.

If $x = x_1 x_2 x_3 \cdots x_n$ is one of the string of length n, then the weight of x is denoted by $wt(x)$ and is defined as

$wt(x) = x_1 + x_2 + \cdots + x_n$. Find the no. of strings of length 10 having even weight (i) How many strings have, 3 0's, from 1's and three 2's (ii) How many strings have at least Eight 1's?

Let $x = x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10}$ ① be a string of length 10. The string in Eq ① has Even weight, then there must be an Even no. of 1's appear in the string.

No. of 1's	No. of strings
$0 \to {}^{10}C_0$	${}^{10}C_0 \times 2^{10}$
$2 \to {}^{10}C_2$	${}^{10}C_2 \times 2^8$
$4 \to {}^{10}C_4$	${}^{10}C_4 \times 2^6$
$6 \to {}^{10}C_6$	${}^{10}C_6 \times 2^4$
$8 \to {}^{10}C_8$	${}^{10}C_8 \times 2^2$
$10 \to {}^{10}C_{10}$	${}^{10}C_{10} \times 2^0$

By the rule of sum, the no. of strings of length 10 that have Even weight

$= [{}^{10}C_0 \times 2^{10}] + [{}^{10}C_2 \times 2^8] + [{}^{10}C_4 \times 2^6]$
$+ [{}^{10}C_6 \times 2^4] + [{}^{10}C_8 \times 2^2] + {}^{10}C_{10}$
$=$

(i) $\frac{10!}{3!\,4!\,3!}$

(ii) $[{}^{10}C_8 \times 2^2] + [{}^{10}C_9 \times 2^1] + [{}^{10}C_{10} \times 2^0]$

(12) For the strings of length 10 made up from the digits 0, 1, 2 ... have weight 3? How many such we

② Find the no. of permutations of the letters of the word MASSASAUGA. In how many of these, all the 3 A's are together? How many of them begin with S?

M - 1, A - 4, S - 3, U - 1, G - 1.

There are 10 letters in the given word.

The no. of arrangements in the given word

$$= \frac{10!}{4!3!}$$

If all four A's are together and treat all four A's as a single letter, then the total no. of letters

$$= 10 - 4 + 1 = 7$$

The no. of arrangements in which all four A's are together $= \frac{7!}{3!}$

S							

If we fix, the first place for S, then we have the following letters

M - 1, A - 4, S - 2, U - 1, G - 1

The no. of arrangements in which they begin with 'S' $= \frac{9!}{4!2!}$

$\therefore$ Total no. of permutations $= \frac{5!\,3!}{}$

Note: In general to move from (x_1, y_1) to (x_2, y_2) in xy-plane (1st quadrant), there are

$$\frac{[(x_2-x_1)+(y_2-y_1)]^2}{(x_2-x_1)!\,(y_2-y_1)!} \text{ paths.}$$

(10) How many distinct paths are there from $(-1,2,0)$ to $(1,3,7)$ in 3-D space, if each move is one of the following:
$H: (x,y,z) \to (x+1, y, z)$: $V: (x,y,z) \to (x, y+1, z)$ $A: (x,y,z) \to (x,y,z+1)$
Generalize the above result.

To move from $(-1,2,0)$ to $(1,3,7)$, we have

$x_2 - x_1 = 2$ H moves
$y_2 - y_1 = 3-2 = 1$ V moves
$z_2 - z_1 = 7$ A moves

$\therefore$ Total permutations $= \frac{10!}{2!\,1!\,7!} = 360$

Generalization :- If $H: (x,y,z) \to (x+m, y, z)$
$V: (x,y,z) \to (x, y+n, z)$
$A: (x,y,z) \to (x, y, z+p)$

then we have m H's, n V's, p A's

$\therefore$ Total no. of perms $= \frac{(m+n+p)!}{m!\,n!\,p!}$.

Circular permutations

Arrangement of objects over a closed path is called a Circular permutation. The no. of circular permutations of n objects is $(n-1)!$

Note ①. Among n objects take one of the object and fix the position. Then remaining $(n-1)$ objects are to be arranged in $(n-1)!$ ways.

②. The no. of circular permutations of r objects from a collection of 'n' is $\frac{1}{r}\,{}^{n}P_{r}$

①. In how many ways can 10 people be arranged over a circular table. If 3 of them insisting on sitting next to each other, how many arrangements are possible?

▶ No. of ways to arrange 10 people over a circular table
$= (10-1)! = 9!$

Now consider three persons together as one person and the three persons can be arranged among themselves in $3!$ ways

Total no. of persons $= 10-3+1 = 8$

No. ways to arrange 3 among 10 people together over a circular table $= (8-1)! \times 3! = 7! \times 3!$

②. In how many ways a group of 4 couples be seated on a round table so that that males and females being seated alternatively?

▶ No. of ways to arrange males $= (4-1)! = 3!$

No. of ways to arrange females $= 4!$

$\therefore$ Total no. of arrangements $= 3! \times 4!$

If couples are to be seated together?

No. of arrangements $= (4-1)! \times 2! = 3! \times 2! = 12$

In how many arrangements the spouses sit opposite each other?

6 Let the 4 couples be denoted by M_1F_1, M_2F_2, M_3F_3, M_4F_4.

Combinations :- combination mainly deals with the ways in which r objects can be selected from a collection of n distinct objects it denoted by ${}^nC_r$ (or) $C(n,r)$ (or) $\binom{n}{r}$ and it is given by $${}^nC_r = \frac{n!}{(n-r)!\,r!} = \frac{{}^nP_r}{r!}$$

Note :① ${}^nC_r$ is the no. of combinations of r objects from a given a collection of 'n' objects. Each combination containing r objects and they can be arranged among themselves in $r!$ ways. Therefore, by the rule of product the no. of permutations $= r! \times {}^nC_r = {}^nP_r$

$$\therefore {}^nC_r = \frac{{}^nP_r}{r!}$$

② ${}^nC_r = {}^nC_{n-r}$ ③ ${}^nC_r + {}^nC_{r-1} = {}^{(n+1)}C_r$

④ In a counting problem, if the order is not important then we use combinations. [My fruit salad is a combination of apples, grapes and bananas, we don care what order the fruits are the

⑤ In a Counting problem, if the order is important then we use permutation (or) rule of product. [The combination to the safe is 472 NOW, we do care about the ord 724 won't work

A student must answer 5 out of 10 questions in a list including at least 3 from the last five questions. In how many ways can a student answer 5 questions?

out of 10 questions 5 must be answered

First five questions

$2 \rightarrow {}^5C_2$

$1 \rightarrow {}^5C_1$

$0 \rightarrow {}^5C_0$

Last five questions

$3 \rightarrow {}^5C_3$

$4 \rightarrow {}^5C_4$

$5 \rightarrow {}^5C_5$

$\therefore$ No. of ways to answer $= {}^5C_2 \times {}^5C_3 + {}^5C_1 \times {}^5C_4 + {}^5C_0 \times {}^5C_5 = 111$

(2) A committee of 8 members is to be formed from 8 teachers and 5 students. In how many ways this can be done if (i) the committee contains exactly 2 students (ii) majority of the teachers (iii) at least 2 students (iv) at most 6 teachers?

No. of Teachers = 8

No. of students = 5

(i) $6 \rightarrow {}^8C_6$ | $2 \rightarrow {}^5C_2$

(ii) $7 \rightarrow {}^8C_7$ | $1 \rightarrow {}^5C_1$

$6 \rightarrow {}^8C_6$ | $2 \rightarrow {}^5C_2$

$5 \rightarrow {}^8C_5$ | $3 \rightarrow {}^5C_3$

(iii) $6 \rightarrow {}^8C_6$ | $2 \rightarrow {}^5C_2$

$5 \rightarrow {}^8C_5$ | $3 \rightarrow {}^5C_2$

$4 \rightarrow {}^8C_4$ | $4 \rightarrow {}^5C_4$

$3 \rightarrow {}^8C_3$ | $5 \rightarrow {}^5C_5$

(iv) $6 \rightarrow {}^8C_6$ | $2 \rightarrow {}^5C_2$

$5 \rightarrow {}^8C_5$ | $3 \rightarrow {}^5C_3$

$4 \rightarrow {}^8C_4$ | $4 \rightarrow {}^5C_4$

$3 \rightarrow {}^8C_3$ | $5 \rightarrow {}^5C_5$

(i) No. of ways to form the committee with exactly two students $= {}^8C_6 \times {}^5C_2$

(ii) No. of ways to form the committee with majority of the teachers $= {}^8C_7 \times {}^5C_1 + {}^8C_6 \times {}^5C_2 + {}^8C_5 \times {}^5C_3 + {}^8C_8 \times {}^5C_0$

(2) A committee of 5 people is to be chosen from a set of 7 women and 4 men. How many ways are there to form the committee if:

(i) The committee consists of 3 women and 2 men?

(ii) The committee can be any size but must have equal number of women and men?

(iii) The committee has 4 people and one of them must be Mr. Rama?

(iv) The committee has 4 people and at least 2 are women?

(v) The committee has 5 people, two of each sex, and Mr. and Mrs. Rama cannot both be on the committee?

(i) $^7C_3 \times {}^4C_2 = 210$.

(ii) $^7C_1 \times {}^4C_1 + {}^7C_2 \times {}^4C_2 + {}^7C_3 \times {}^4C_3 + {}^7C_4 \times {}^4C_4$

Women
$1 \to {}^7C_1$
$2 \to {}^7C_2$
$3 \to {}^7C_3$
$4 \to {}^7C_4$

Women
$1 \to {}^4C_1$
$2 \to {}^4C_2$
$3 \to {}^4C_3$
$4 \to {}^4C_4$

(iii) $^{10}C_3$

(iv)

W	M
$2 \to {}^7C_2$	$4 \to {}^4C_2$
$3 \to {}^7C_3$	$1 \to {}^4C_1$
$4 \to {}^7C_4$	$0 \to {}^4C_0$

W = 7	M = 4
$2 \to {}^7C_2$	$2 \to {}^4C_2$
$3 \to {}^7C_2$	$1 \to {}^4C_2$
$4 \to {}^7C_4$	$0 \to {}^4C_0$

$= {}^7C_2 \times {}^4C_2 + {}^7C_3 \times {}^4C_1 + {}^7C_4 \times {}^4C_0 = 301$

(v) (i) Mrs. Rama is on the committee and Mr. Rama is not. Then '1' more women must be chosen from the remaining 6 women and 2 more men must be chosen from the remaining 3 men. This can be done in $= {}^6C_1 \times {}^3C_2 = 18$ ways.

(ii) Mrs. Rama is off and Mr. Rama is on, which by a similar argument yields $= {}^6C_2 \times {}^3C_1 = 45$ ways

(iii) neither of them is on the committee, we have

$= {}^6C_2 \times {}^3C_2 = 45$.

$\therefore$ The total no. of ways $= 18 + 45 + 45 = 108$.

$^7C_2 \times {}^4C_2 =$
$^6C_1 \times {}^3C_1 =$

How many ways can a committee be formed from six men and six women with

(i) At least two men and at least twice as many women as men?

(ii). Between 3 and 5 people and Mrs. Bularya if excluded

M=4

$2 \rightarrow {}^4C_2$

$3 \rightarrow {}^4C_3$

W=6

$4 \rightarrow {}^6C_4$; $5 \rightarrow {}^6C_5$; $6 \rightarrow {}^6C_6$

${}^4C_2 \times [{}^6C_4 \times {}^6C_5 \times {}^6C_6] + {}^4C_3 \times {}^6C_6$

(ii) M=4

W=5 Excluded Ishwar

${}^9C_3 + {}^9C_4 + {}^9C_5$

5. A committee of 8 is to be formed from 16 men and 10 women. In how many ways can the committee be formed if (i) there are no restrictions (ii) there must be 4 men and 4 women (iii) there should be even no. of women (iv) more women than men (v) at least 6 men

(i) ${}^{26}C_8$

(ii) ${}^{16}C_4 + {}^{10}C_4 \cdot {}^{10}C_4 \times {}^{16}C_4 + {}^{10}C_6 \times {}^{16}C_2 + {}^{10}C_8$

(iii) ${}^{10}C_0 \times {}^{16}C_8 + {}^{10}C_2 \times {}^{16}C_6$

(iv)

men = 16

$0 \rightarrow {}^{16}C_0$

$1 \rightarrow {}^{16}C_1$

$2 \rightarrow {}^{16}C_2$

$3 \rightarrow {}^{16}C_3$

$4 \rightarrow {}^{16}C_4$

women = 10

$8 \rightarrow {}^{10}C_8$

$7 \rightarrow {}^{10}C_7$

$6 \rightarrow {}^{10}C_6$

$5 \rightarrow {}^{10}C_5$

$4 \rightarrow {}^{10}C_4$

(v)

$6 \rightarrow {}^{16}C_6$

$7 \rightarrow {}^{16}C_7$

$8 \rightarrow {}^{16}C_8$

$2 \rightarrow {}^{10}C_2$

$1 \rightarrow {}^{10}C_1$

$0 \rightarrow {}^{10}C_0$

In how many ways can a student answer a
7 questions from a set of 10 questions such that the
student answer at least 3 questions from the first 5 questions
out of 10 questions 7 must be answered

First Five questions

$3 \rightarrow {}^5C_3$

$4 \rightarrow {}^5C_4$

$5 \rightarrow {}^5C_5$

Last Five questions

$4 \rightarrow {}^5C_4$

$3 \rightarrow {}^5C_3$

$2 \rightarrow {}^5C_2$

$\therefore$ No. of ways to answer $= {}^5C_3 \times {}^5C_4 + {}^5C_4 \times {}^5C_3 + {}^5C_5 \times {}^5C_2 = 110$

(15). How many arrangements of the letters in MISSISSIPPI have no consecutive S's.

$- M - I - I - I - P - P - I -$

There are $\frac{7!}{4!2!}$ arrangements with out 'S'

There are 8 possible places for S and that can be selected in ${}^8C_4$ ways.

Thus, there are $\frac{7!}{4!2!} \times {}^8C_4$ arrangements are available with no consecutive S's.

(16). How many arrangements of EFFLORESCENCE has consecutive C's and consecutive F's but no consecutive E's?

We have : E-4, F-2, C-2, L-1, O-1, R-1, N-1, S-1.

If 2 C's and 2 F's are together, then we have the following letters

E-4, F-1, C-1, L-1, O-1, R-1, N-1, S-1

with out 4 E's, we have the following :

$- F - C - L - O - R - N - S -$

There are 7! arrangements without E's

There are 8 possible places for E's and that can be selected in ${}^8C_4$ ways.

Thus, there are $7! \times {}^8C_4$ arrangements are there with consecutive C's and consecutive F's but no consecutive E's.

(17). Find the no. of arrangements of all the letters in TALLAHASSEE. How many of these arrangements have no adjacent A's?

No. of letters in the given word = 11

A-3, L-2, S-2, E-2, T-1, H-1

∴ The no. of arrangements of the letters in the given word is $= \frac{11!}{3!\,2!\,2!\,2!}$

- T - L - L - H - S - S - E - E -

If we discard the all three A's, then we have

L-2, S-2, E-2, T-1, H-1

and these can be arranged in $\frac{8!}{2!\,2!\,2!}$

There are '9' possible places for the 3A's and these can be selected in ${}^9C_3$ ways.

∴ The no. of arrangements having no adjacent A's $= \frac{8!}{2! \times 2! \times 2!} \times {}^9C_3$

Combinations with repeatations

Suppose, we wish to select with repeatation, a collection of r objects from a set of n distinct objects.

The no. of such selections is given by

$$C(n+r-1, r) = \frac{(n+r-1)!}{(n-1)!\, r!} = C(r+n-1, n-1)$$

In other words,

$C(n+r-1, r) = C(r+n-1, n-1)$ represents the no. of combinations of 'n' disjoint objects, taken r at a time, with repeatations allowed.

The following are the other interpretations of this number.

(1) $C(n+r-1, r) = C(r+n-1, n-1)$ represents the no. of ways in which 'r' identical objects can be distributed among 'n' distinct containers.

(2) $C(n+r-1, r) = C(r+n-1, n-1)$ represents the no. of non-negative integer solutions of the equation

$x_1 + x_2 + \cdots + x_n = r$, each $x_i \geq 0$, $1 \leq i \leq n$.

(1) In how many ways can we distribute 10 identical marbels among 6 distinct containers.

Here $n = 6$, $r = 10$

$$C(n+r-1, r) = C(10+6-1, 10) = \frac{15!}{5!\,10!} = 3003.$$

(2) Find the number of non-negative integer solutions of the Equation $x_1 + x_2 + x_3 + x_4 + x_5 = 8$.

Here $n = 5$ and $r = 8$, all x_i's ≥ 0, $1 \leq i \leq 5$.

$$C(n+r-1, r) = C(5+8-1, 8) = \frac{12!}{4!\,8!} = 495.$$

③ Find the no. of non-negative integer solutions of the inequality $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 < 10$, where $x_i \geq 0$, $1 \leq$

Let $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 10$, Here

$x_i \geq 0$, $1 \leq i \leq 6$ and $x_7 \geq 1$.

This $(\Rightarrow)$ $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + 1 = 10$

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 9$

Now Each $x_i \geq 0$, $1 \leq i \leq 7$

Here $n = 7$ $\therefore$ $r = 9$.

The no. of non-negative integer solutions

$= c(n+r-1, r) = c(9+7-1, 9) = c(15, 9) = \frac{15!}{6!\,9!}$

④ Find the no. of +ve integral solutions of the Equation $x_1 + x_2 + x_3 + x_4 + x_5 = 30$, where $x_1 \geq 2$, $x_2 \geq 3$, $x_3 \geq 4$, $x_4 \geq 2$ and $x_5 \geq 0$.

$(x_1 + 2) + (x_2 + 3) + (x_3 + 4) + (x_4 + 2) + x_5 = 30$

$x_1 + x_2 + x_3 + x_4 + x_5 = 19$

Each $x_i \geq 0$, $1 \leq i \leq 5$.

$n = 5$ $r = 19$

$\therefore$ The no. of non-negative integer solutions

$c(5+19-1, 19) = c(23, 4) = \frac{23!}{19!\,4!}$

⑤ Find the no. of +ve integer solutions of the Equation

$x_1 + x_2 + x_3 = 17$.

Given that $x_1 + x_2 + x_3 = 17$

where $x_1 \geq 1$, $x_2 \geq 1$, $x_3 \geq 1$.

$(x_1 + 1) + (x_2 + 1) + (x_3 + 1) = 17$

$x_1 + x_2 + x_3 = 14 - (1)$

$r = 14$; $n = 3$

The no. of non-negative integer solutions of the Eq (1) is

In how many ways, can we arrange 24 different books on 4 book shelves, so that there is at least one book on each shelf.

Let x_1, x_2, x_3 and x_4 be denote 4 book shelves.

Then, we have

$x_1 + x_2 + x_3 + x_4 = 24$, where each $x_i \geq 1$, $1 \leq i \leq 4$.

$\Rightarrow (x_1+1) + (x_2+1) + (x_3+1) + (x_4+1) = 24$

$\Rightarrow x_1 + x_2 + x_3 + x_4 = 20$, can $x_i \geq 0$, $1 \leq i \leq 4$.

Here $n = 4$; $r = 20$

The required number is $= c(n+r-1, r) = c(4+20-1, 20)$

$= c(23, 20)$.

7. A message is made up of 12 different symbols and is to be transmitted through a communication channel. In addition to 12 symbols, the transmitter will also send a total of 45 spaces b/w the symbols with at least 3 spaces b/w each pair of consecutive symbols. In how many ways can transmitter send such a message.

A B C D E F G H I J K L

From 12 symbols, we can form 12! messages. For each of these arrangements, there are 11 positions b/w 12 different symbols. Since there must be 3 space b/w successive symbols, 33 of the 45 spaces will be used up. The remaining 12 spaces are to be accomodated in 11 positions. This can be done in

$c(11+12-1, 12) = c(22, 12)$ ways.

Consequently, by the product rule, the required number is $= 12! \times c(22, 12) = \frac{22!}{10!}$

nomial Theorem

If x and y are any two variables and 'n' is a +ve integer

then $[x+y]^n = \sum_{r=0}^{n} {}^nC_r\, x^r\, y^{n-r}$.

Note: ① ${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n = 2^n$

② ${}^nC_0 - {}^nC_1 + {}^nC_2 - \cdots + (-1)^n {}^nC_n = 0$.

Multinomial Theorem

For the +ve integers n and k, the coefficient of $x_1^{n_1} \times x_2^{n_2} \times \cdots \times x_k^{n_k}$ in the Expansion of $[x_1 + x_2 + \cdots + x_k]^n$ is

$$\frac{n!}{n_1!\, n_2! \times \cdots \times n_k!}$$

where each n_i is a non-negative integer $\le n$ and $n_1 + n_2 + \cdots + n_k = n$.

Note:- -11- in $[a_1 x_1 + a_2 x_2 + \cdots + a_k x_k]^n$ is

$$\frac{n!}{n_1!\, n_2! \times \cdots \times n_k!} \times \left[a_1^{n_1} \times a_2^{n_2} \times \cdots \times a_k^{n_k}\right]$$

② Another way of stating the multinomial theorem is, when 'n' is a +ve integer, the general term in the expansion of $[x_1 + x_2 + \cdots + x_k]^n$ is $\frac{n!}{n_1! \times n_2! \times \cdots \times n_k!} x_1^{n_1} \times x_2^{n_2} \times \cdots \times x_k^{n_k}$

where each n_i is a non-negative integer $\le n$ and

$n_1 + n_2 + \cdots + n_k = n$.

∴ The Expression $\frac{n!}{n_1! \times n_2! \times \cdots \times n_k!}$ is called a

multinomial coefficient.

① Find the coefficient of x^3y^9 in the expansion of $(x+y)^{12}$.

We have $(x+y)^n = \sum_{r=0}^{n} {}^nC_r\, x^r\, y^{n-r}$

$[x+y]^{12} = \sum_{r=0}^{12} {}^{12}C_r\, x^r\, y^{12-r}$

In the expansion, the coefficient of x^3y^9 (which corresponds to $r=3$) is ${}^{12}C_3\, x^3 y^9$

$\therefore$ The coefficient of x^3y^9 is ${}^{12}C_3$

② Find the coefficient of x^9y^3 in the Expansion of $[2x-3y]^{12}$

We have $[x+y]^n = \sum_{r=0}^{n} {}^nC_r\, x^r\, y^{n-r}$

$[2x-3y]^{12} = [2x+(-3y)]^{12} = \sum_{r=0}^{12} {}^{12}C_r\, (2x)^r\, (-3y)^{12-r}$

In the Expansion, the coefficient of x^9y^3 (which corresponds to $r=9$) is

${}^{12}C_9\, (2x)^9\, (-3y)^3 = {}^{12}C_9\, 2^9\, x^9\, (-3)^3\, y^3$.

$\therefore$ The coefficient of x^9y^3 is ${}^{12}C_9\, 2^9\, (-3)^3$.

③ Find the coefficient of a^5b^2 in the expansion of $[2a-3b]^7$.

$[2a+(-3b)]^7 = \sum_{r=0}^{7} {}^7C_r\, (2a)^r\, (-3b)^{7-r}$

In the expansion, the coefficient of a^5b^2 (which corresponds to $r=5$) is

${}^7C_5\, (2a)^5\, (-3b)^2 = {}^7C_5\, 2^5\, a^5\, (-3)^2\, b^2$

$\therefore$ The coefficient of a^5b^2 is $= {}^7C_5\, 2^5 \times 3^2$.

④ Find the coefficient of x^5y^6 in $\left[\frac{3x}{2} - \frac{5y}{3}\right]^{11}$.

$\left[\frac{3x}{2} + \left(-\frac{5y}{3}\right)\right]^{11} = \sum_{r=0}^{11} {}^{11}C_r \left[\frac{3x}{2}\right]^r \left(-\frac{5y}{3}\right)^{11-r}$

In the Expansion, the coefficient of x^5y^6 which corresponds to $r=5$ and is written as

${}^{11}C_5 \left(\frac{3x}{2}\right)^5 \left(-\frac{5y}{3}\right)^{11-5} = {}^{11}C_5 \left(\frac{3}{2}\right)^5 x^5 \left(-\frac{5}{3}\right)^6 y^6 = {}^{11}C_5\, \frac{5^6}{3 \times 2^5}\, x^5y^6$.

we have $\binom{\quad}{5,3,2,2}$ $= 5!$

(8) Find the multinomial coefficient of $x^2 y^3 z^3$ in the expansion of $[x+y+z]^6$.

From the multinomial theorem, the coefficient of $x^2 y^3 z^3$ is

$$\frac{6!}{2!1!3!} = \frac{6\times5\times4\times3\times2\times1}{2\times3!} = 60.$$

(9) -11 ——— $x^2 y^2 z^3$ in -11 ——— $[x+y+z]^7$.

-11 ——— $\frac{7!}{2!2!3!} = 210.$

(10) -11 ——— xyz^5 in -11 ——— $[x+y+z]^7$.

-11 ——— $\frac{7!}{5!} = 42.$

(11) -11 ——— $x^3 z^4$ in -11 ——— $[x+y+z]^7$.

-11 ——— $\frac{7!}{3!0!\times4!} = 35.$

11. 11. 2!

(16) Find the multinomial coefficient of $x^2 y^3 z^4 t^3$ in $[2x - y - z - 2t + 3]^{16}$.

The general term in the expansion of $[2x - y - z - 2t + 3]^{16}$ is

$$\frac{16!}{n_1!\, n_2!\, n_3!\, n_4!\, n_5!} (2x)^{n_1} (-y)^{n_2} (-z)^{n_3} (-2t)^{n_4} (3)^{n_5}$$

we have $n_1 = 2$; $n_2 = 3$; $n_3 = 4$; $n_4 = 3$

we have $n_1 + n_2 + n_3 + n_4 + n_5 = 16$

$\therefore n_5 = 16 - 12 = 4$.

$\therefore$ The coefficient of $x^2 y^3 z^4 t^3$ is

$$\frac{16!}{2!\,3!\,4!\,3!\,4!} 2^2 \times (-1)^3 \times (-1)^4 \times (-2)^3 \times (3)^4$$

16

Find the multinomial coefficient of $xy^2z^3t^2$ in $[3x-2y+2z-4t-15]^{14}$.

$$\frac{14!}{1!\,2!\,3!\,2!\,6!}(3)^1\times(-2)^2\times(2)^3\times(-4)^2\times(-15)^6.$$

Find the coefficient of $w^2x^2y^2z^2$ in the expansion of

(i) $[w+x+y+z+1]^{10}$ (ii). $[2w-x+3y+z-2]^{12}$

(iii). $[v+w-2x+y+5z+3]^{12}$.

(i) $\frac{10!}{2!\,2!\,2!\,2!\,2!}$ (ii). $\frac{12!}{2!\,2!\,2!\,2!\,4!}2^2\times(-1)^2\times(3)^2\times(1)^2\times(-2)^4$

(iii). $\frac{12!}{0!\,2!\,2!\,2!\,2!\,4!}(1)^2\times(-2)^2\times(1)^2\times(5)^2\times(3)^4$

(20). Determine the coefficient of

(i) xyz^2 in $[x+y+z]^4$ (iii) xyz^2 in $[2x-y-3]^4$

(ii) xyz^2 in $[w+x+y+z]^4$ (iv) $w^3x^2y^2z$ in $[2w-x+3y-2z]^8$.

(i) $\frac{4!}{1!\,1!\,2!}$ (iii). $\frac{4!}{1!\,1!\,2!}(2)^1\times(-1)^1\times(-1)^2=-24$

(ii). $\frac{4!}{0!\,1!\,1!\,2!}\times(1)^1\times(1)^1\times(1)^2=12$. (iv). $\frac{8!}{3!\times2!\times1!\times2!}2^3\times(-1)^2\times(3)^1\times(-2)^2$

(21). Find the multinomial coefficient of xyz^{-2} in the Expansion of $[x-2y+z^{-1}]^4$.

Put $z^{-1}=t \Rightarrow [z^{-1}]^2=t^2$

$\therefore z^{-2}=t^2$.

Now, we have to find the coefficient of xyt^2 in the Expansion of $[x-2y+t]^4$.

$\therefore$ The coefficient of xyt^2 is $= \frac{4!}{1!\,1!\,2!}(1)^1\times(-2)^1\times(1)^2=-2$